



**PEOPLES INFORMATION TECHNOLOGY PROGRAM**  
**QUEST CAMPUS, NAWABSHAH**



**DATABASE MANAGEMENT SYSTEMS**

**FINAL EXAMINATION**

**DATE: January 10, 2025,**

**Time Allowed: 2 hours**

**Maximum Marks: 50**

**Part A: Multiple-Choice Questions (60 Marks)**

**Each question carries 2 marks. Choose the correct option.**

1. Which SQL statement is used to retrieve data from a database?  
a) DELETE                      b) SELECT                      c) INSERT                      d) CREATE
2. What does CRUD stand for in database operations?  
a) Create, Read, Update, Delete                      b) Copy, Replace, Update, Delete  
c) Create, Replace, Update, Delete                      d) None of the above
3. Which of the following is NOT a normalization form?  
a) First Normal Form (1NF)                      b) Second Normal Form (2NF)  
c) Fifth Normal Form (5NF)                      d) Combined Normal Form (CNF)
4. Which MySQL data type is used to store large text?  
a) TEXT                      b) VARCHAR                      c) INT                      d) CHAR
5. Which of the following MongoDB commands is used to insert a document?  
a) db.insert()                      b) db.collection.insertOne()                      c) db.add()                      d) db.create()
6. In MySQL, what is a view?  
a) A virtual table                      b) A physical copy of a table  
c) A data type                      d) None of the above
7. Which of the following is NOT a DML command?  
a) INSERT                      b) UPDATE                      c) DELETE                      d) CREATE
8. In MongoDB, which of the following is used to perform an aggregation operation?  
a) db.collection.aggregate()                      b) db.collection.group()  
c) db.collection.query()                      d) db.collection.filter()
9. Which command in MySQL creates a database?  
a) CREATE DATABASE db\_name;                      b) NEW DATABASE db\_name;  
c) INSERT DATABASE db\_name;                      d) INIT DATABASE db\_name;
10. What does DDL stand for?  
a) Data Definition Language                      b) Data Description Language  
c) Data Directive Language                      d) Data Declarative Language
11. Which aggregate function is used to find the maximum value in a column?  
a) MIN()                      b) MAX()  
c) AVG()                      d) SUM()
12. Which is NOT an aggregate function in SQL?  
a) COUNT()                      b) SUM()  
c) AVG()                      d) STRING()

13. In MongoDB, what does the find() method return?
- a) A list of documents
  - b) A single document
  - c) A table
  - d) An array
14. Which SQL clause is used to group rows that have the same values?
- a) WHERE
  - b) GROUP BY
  - c) HAVING
  - d) ORDER BY
15. How does MongoDB store data?
- a) Tables
  - b) JSON-like documents
  - c) XML files
  - d) Rows and columns
16. Which of the following is a valid auto-backup tool in Windows 11?
- a) Task Manager
  - b) Task Scheduler
  - c) AutoSave
  - d) Data Manager
17. In SQL, which command is used to modify an existing table?
- a) ALTER
  - b) MODIFY
  - c) UPDATE
  - d) CHANGE
18. Which key is used to uniquely identify a row in a table?
- a) Foreign Key
  - b) Primary Key
  - c) Unique Key
  - d) Composite Key
19. Which MongoDB operator is used for filtering documents?
- a) \$filter
  - b) \$match
  - c) \$find
  - d) \$search
20. Which of the following is a feature of MySQL?
- a) Open-source
  - b) Supports relational databases
  - c) Platform independent
  - d) All of the above
21. Which SQL function calculates the total number of records?
- a) COUNT()
  - b) SUM()
  - c) TOTAL()
  - d) CALC()
22. What is the output of the following MongoDB command? db.collection.find().limit(5)
- a) All documents
  - b) First 5 documents
  - c) Last 5 documents
  - d) None
23. What does the AUTO\_INCREMENT attribute in MySQL do?
- a) Automatically updates a column
  - b) Automatically increments a column value
  - c) Automatically creates a new row
  - d) None of the above
24. Which normalization form ensures no partial dependency?
- a) 1NF
  - b) 2NF
  - c) 3NF
  - d) BCNF
25. In MongoDB, how can you drop a collection?
- a) db.collection.drop()
  - b) db.collection.remove()
  - c) db.collection.delete()
  - d) db.collection.clear()

26. Which command is used to create a new user in MySQL?
- a) CREATE USER
  - b) ADD USER
  - c) INSERT USER
  - d) NEW USER
27. Which tool in Windows 11 can be used for scheduling auto-backups?
- a) Task Scheduler
  - b) File Explorer
  - c) Backup Manager
  - d) Control Panel
28. Which MongoDB method updates multiple documents?
- a) update()
  - b) updateMany()
  - c) updateOne()
  - d) modify()
29. Which SQL command deletes a table?
- a) DROP TABLE
  - b) DELETE TABLE
  - c) REMOVE TABLE
  - d) CLEAR TABLE
30. Which of the following is a DDL command?
- a) SELECT
  - b) CREATE
  - c) DELETE
  - d) UPDATE
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**Part B: Descriptive Questions (40 Marks)**

1. Explain the key differences between MySQL and MongoDB. Provide at least three points for comparison. (20 Marks)
2. Explain the ACID properties in the context of database transactions. Provide examples to illustrate how each property ensures data integrity. (20 Marks)